Deep Learning and Its Applications (CO3133) – Assignment 1

Foundations of Deep Learning Pipelines and Architectures

From Linear Models to Modern Sequence Models: A Comparative Study for Image Classification

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Problem Formulation

- **Task:** single-label image classification, one prediction per image.
- **Input:** a grayscale image $x \in \mathbb{R}^{1\times 28\times 28}$.
- **Output:** raw logits $z = f_\theta(x) \in \mathbb{R}^{10}$, one score per class; the prediction is $\hat{y} = \arg\max_c z_c$.
- **Research question:** with the data, the split and the training protocol fixed, how does the model family, and the way it reads the image, change macro-F1, accuracy and cost?
- **Datasets:**
  - Fashion-MNIST [1] for every reported result.
  - MNIST for debugging the pipeline only; no MNIST number is reported.
- **Main difficulty:** four upper-body classes share one silhouette at 28×28 pixels and without colour.

Fashion-MNIST

- **Source:** Zalando product photos, converted to grayscale and resized [1].
- **Size:** 70,000 images; 60,000 official training, 10,000 official test.
- **Format:** 28×28, one channel, `uint8` in $[0, 255]$; about 30 MB compressed.
- **Classes:** T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot.
- **License:** MIT.
- **Balance:** every class holds exactly 10% of each split (Figure 1), so there is no class imbalance to correct.

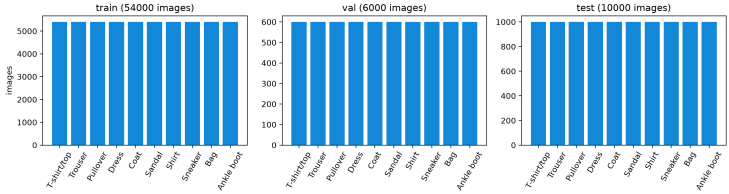


Figure 1: Images per class in the train, validation and test splits.

Representative Samples

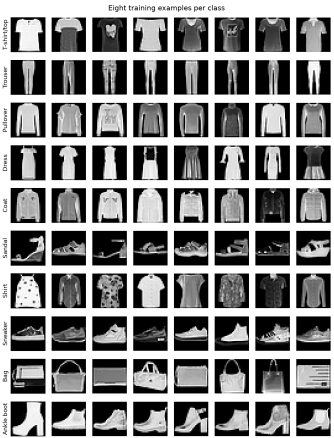


Figure 2: Eight training images per class.

- **Distinct shapes:** Trouser, Bag and the three footwear classes have outlines no other class shares (Figure 2).
- **Shared silhouette:** T-shirt/top, Pullover, Coat and Shirt differ mainly in sleeve length, collar, buttons or zip, details that cover only a few pixels.
- **Variation within a class:** prints, brightness and cut change from image to image.
- **Fixed framing:** every garment is centred on a black background.

Class Overlap Before Training

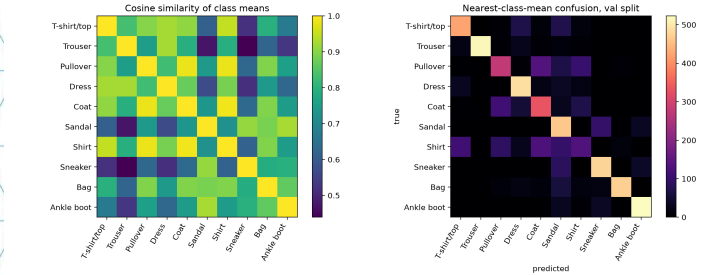
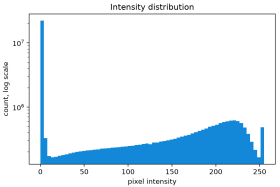


Figure 3: Left: cosine similarity between class mean images. Right: confusion matrix of a nearest-class-mean classifier on the validation split.

- **Similar means:** cosine similarity exceeds 0.98 for Coat/Shirt, Pullover/Coat and Pullover/Shirt.
- **Floor without training:** assigning each image to the closest class mean reaches 68.4% validation accuracy (Figure 3).
- **Uneven classes:** Shirt recall is 0.220, against 0.872 for Ankle boot.

Pixel Statistics and Spatial Structure



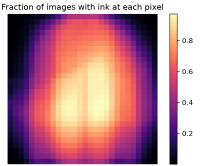


Figure 4: Pixel intensity histogram; share of images with a non-zero value at each pixel.

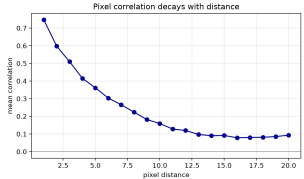


Figure 5: Mean pixel correlation against distance.

- **Sparse input:** half of all training pixels are exactly 0, and the garment occupies the centre (Figure 4), so many of the 784 flattened features carry little information.
- **Redundancy:** 10 principal components explain 72.1% of pixel variance; 100 explain 91.4%.
- **Local structure:** pixel correlation falls from 0.747 at 1 pixel to 0.159 at 10 and 0.093 at 20 (Figure 5). A convolution uses this structure; flattening removes it.

Three Sequence Views of One Image

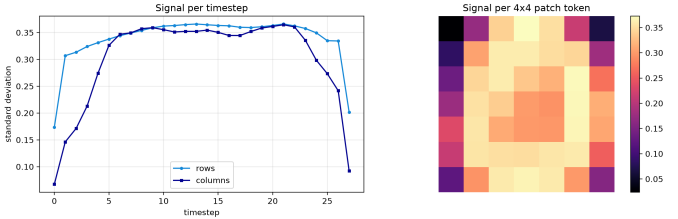


Figure 6: Standard deviation of pixel values per timestep for row and column sequences (left) and per 4×4 patch token (right).

- **Rows or columns:** 28 timesteps of 28 features. **Patches of 4×4:** 49 tokens of 16 features.
- **Where the signal sits:** no row falls below a quarter of the peak standard deviation, against one column and three patch tokens (Figure 6). Patches gather the background into a few corner tokens; rows and columns spread it over every step.

What the Data Implies for the Method

- **Balanced classes:** accuracy and macro-F1 are both valid; macro-F1 is primary because it weights a weak class such as Shirt as much as a strong one.
- **Upper-body overlap:** the error analysis should focus on T-shirt/top, Pullover, Coat and Shirt, and the nearest-class-mean result (68.4%) is the floor every trained model must clear.
- **Local correlation:** a model with a spatial prior, the CNN, is expected to lead at comparable capacity.
- **Uneven signal per token:** patch tokens are expected to beat row and column tokens for the recurrent and Transformer models.
- **Centred garments with small shifts:** random crops of ± 2 pixels and horizontal flips produce plausible training images without changing the label.

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Pipeline Overview

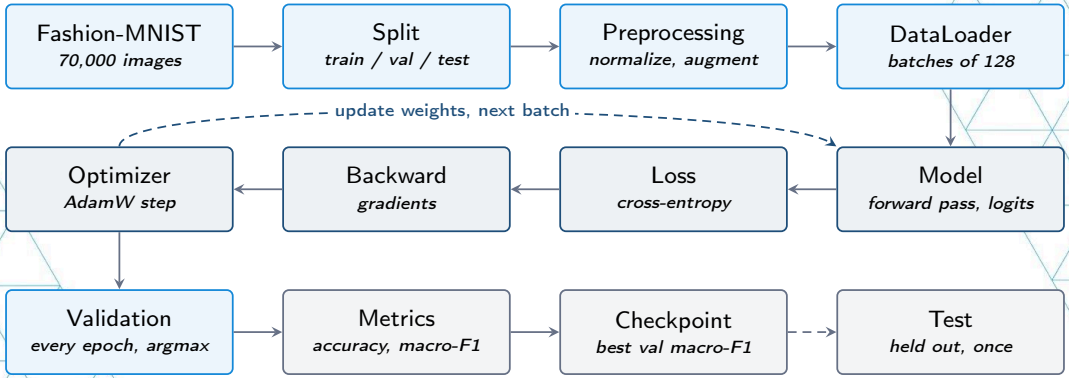


Figure 7: From the dataset to the selected checkpoint; the same pipeline trains all five models.

- **Shared path:** only the *Model* box changes between experiments (Figure 7).
- **Held-out test:** the test split is evaluated once, after model selection on validation macro-F1.

Preprocessing and Batches

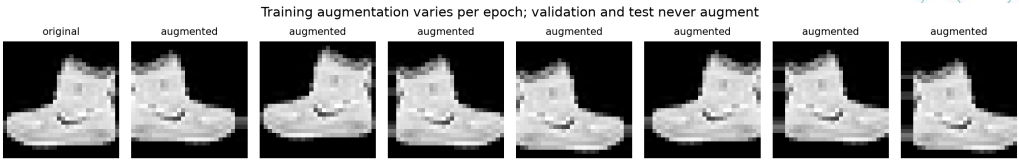


Figure 8: One training image and seven augmented versions of it.

- **Normalization:** $\tilde{x} = (x/255 - 0.2858)/0.3529$, with the mean and standard deviation computed on the training split only.
- **Augmentation, training loader only:** random crop with 2-pixel edge padding, then horizontal flip with probability 0.5 (Figure 8). Validation and test images are never augmented.
- **Batch:** images $(128, 1, 28, 28)$ `float32`, values in $[-0.81, 2.02]$; labels $(128,)$ `int64`. Every model receives this same batch.

Shared Design of the Five Models

- **One output contract:** every model maps a batch to raw logits of shape $(N, 10)$. No model applies softmax; the loss does it:

$$\mathcal{L}(z, y) = -\log \frac{\exp(z_y)}{\sum_{c=1}^{10} \exp(z_c)}$$

- **Registry:** models are registered by name and built from a YAML configuration, so the training loop has no model-specific code.
- **Sequence input:** the recurrent and Transformer models receive the same image batch and convert it into rows, columns or patches themselves.
- **Self-implemented:** the five architectures, the data split, the training and evaluation loops, and run tracking.
- **From libraries:** PyTorch [2] layers (`Linear`, `Conv2d`, `GRU`, `LSTM`, `TransformerEncoderLayer`), optimizer and scheduler; torchvision for download and transforms; scikit-learn for the stratified split and the metrics. No pretrained weights are used.

Model 1: Linear Classifier (Baseline)

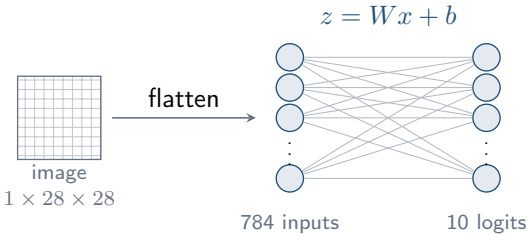


Figure 9: The linear classifier: one affine layer from flattened pixels to class scores.

- **Computation:** $x \in \mathbb{R}^{784}$, $W \in \mathbb{R}^{10\times784}$, $b \in \mathbb{R}^{10}$ (Figure 9); 7,850 parameters.
- **What it can represent:** each class score is a weighted sum of pixels, so the decision boundaries between classes are hyperplanes in pixel space.
- **Role:** the capacity floor of the comparison; every other model must improve on it.

Model 2: Multilayer Perceptron (MLP)

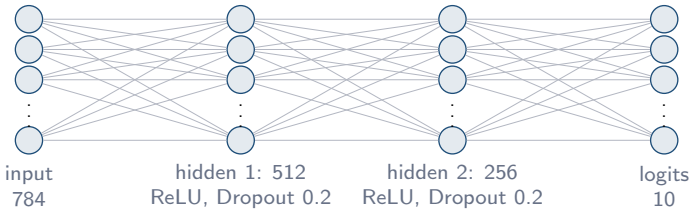


Figure 10: The MLP: two fully connected hidden layers between the flattened image and the logits.

- **Computation:** $h_1 = \mathrm{Dropout}(\mathrm{ReLU}(W_1 x + b_1))$, $h_2 = \mathrm{Dropout}(\mathrm{ReLU}(W_2 h_1 + b_2))$, $z = W_3 h_2 + b_3$ (Figure 10); 535,818 parameters.
- **Activation:** $\mathrm{ReLU}(a) = \max(0, a)$ makes the model non-linear, and its gradient does not shrink for positive inputs.
- **Regularization:** dropout [3] sets 20% of hidden units to zero during training; weight decay 0.05 comes from the shared optimizer.
- **Limit:** like the linear model, it reads a flat vector and has no notion of which pixels are neighbours.

Model 3: Convolutional Neural Network (CNN)

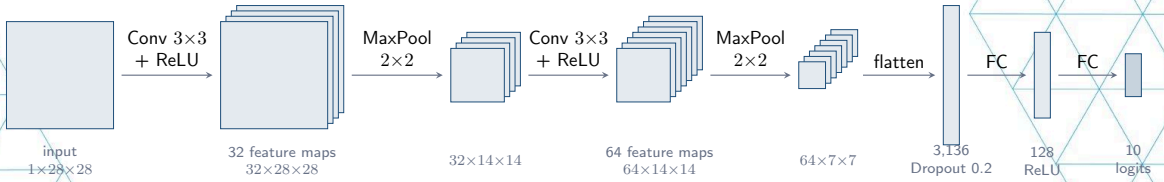


Figure 11: The CNN: two convolution stages, then a fully connected head; 421,642 parameters.

- **Convolution:** each 3×3 filter is applied at every position with the same weights; padding 1 keeps the 28×28 size. One filter produces one feature map, which records where that filter responds (Figure 11).
- **Pooling:** 2×2 max pooling keeps the largest response in each window. It halves the map size and makes the output less sensitive to shifts of one pixel.
- **Design:** our own architecture after LeNet [4], two stages of 32 and 64 filters, trained from scratch.

Model 4: Recurrent Network (GRU)

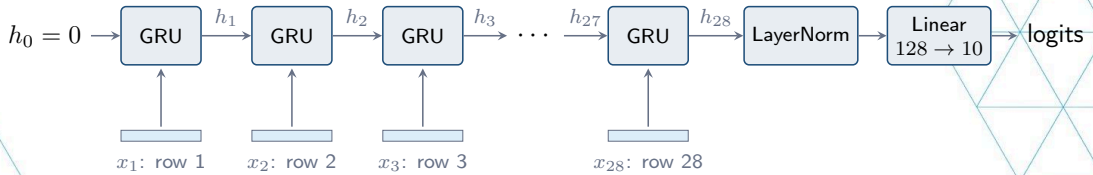


Figure 12: The recurrent classifier reads the image one row per timestep; 62,218 parameters with a GRU.

- **Timestep and input size:** one timestep is one image row, read top to bottom, so the sequence has 28 steps of 28 features (Figure 12). Columns and 4×4 patches are the alternatives tested in E2.
- **Hidden representation:** $h_t = \mathrm{GRU}(x_t, h_{t-1}) \in \mathbb{R}^{128}$ summarizes the rows read so far; the last state h_{28} is classified.
- **Gates:** the GRU [5] uses an update and a reset gate to decide how much of h_{t-1} to keep. The LSTM [6] adds a separate cell state and a third gate (82,442 parameters); E3 compares the two.

Model 5: Transformer Encoder

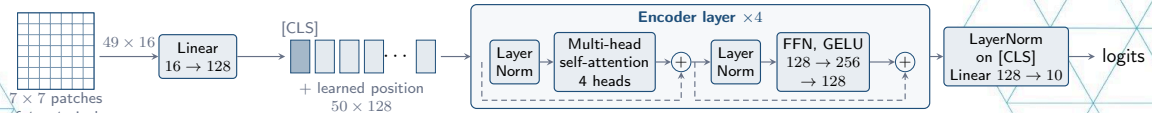


Figure 13: The Transformer classifier over 4×4 patch tokens; 540,170 parameters.

- **Tokens:** each patch is flattened to 16 values and projected to $d = 128$; a learned [CLS] token is prepended, and its final state is classified (Figure 13), as in ViT [7].
- **Positional encoding:** attention ignores token order, so one learned vector per position is added to each token.
- **Attention [8]:** queries, keys and values are linear projections of the same tokens H; each of the 4 heads computes $\mathrm{softmax}(QK^\top/\sqrt{32})\,V$, so every token can use information from every other token in one layer. Input and output are both $(N, 50, 128)$.

The Five Models Side by Side

Table 1: Input view, built-in assumption and size of each model, at the configurations in `configs/`.

Model	Input view	Built-in assumption (inductive bias)	Parameters
Linear	784 pixels	class scores are linear in the pixels	7,850
MLP	784 pixels	none about pixel layout	535,818
CNN	28×28 grid	nearby pixels matter together; same filter everywhere	421,642
GRU	28 rows of 28	the image is an ordered sequence of rows	62,218
Transformer	49 patches of 16	few; positions are learned, all tokens interact	540,170

- **No spatial prior:** Linear and MLP would learn equally well if every image had its pixels shuffled by the same fixed permutation; the CNN and the patch Transformer would not.
- **Unequal size:** parameter counts differ by a factor of 69 (Table 1), so results are read together with cost.

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Data Split and Leakage Control

Table 2: The split shared by every model, stratified by class with seed 0.

Subset	Images	Per class	Source
Train	54,000	5,400	official training set, 90%
Validation	6,000	600	official training set, 10%
Test	10,000	1,000	official test set

- **Fixed partition:** the split indices are committed in `splits/fashion-mnist-seed0.json` and read by every run (Table 2), instead of being regenerated from a seed that depends on library versions.
- **No information from validation or test in training:** normalization statistics come from the training split only, and augmentation is applied to the training loader only.
- **Single test evaluation:** checkpoints and configurations are chosen on validation; the test split is evaluated once, after model selection concludes.

Training Protocol

Table 3: Training settings shared by all five models.

Setting	Value
Loss	cross-entropy on raw logits, no label smoothing
Optimizer	AdamW [9], learning rate 3×10^{-4}, weight decay 0.05
Batch size	128
Schedule	cosine annealing [10] over 30 epochs
Early stopping	after 5 epochs without improvement in validation macro-F1
Checkpoint	epoch with the highest validation macro-F1
Augmentation	random crop with 2-pixel edge padding, horizontal flip
Seeds	0, 1, 2 for every reported configuration

- **Environment:** one Apple M4 Pro (14 cores, 48 GB) for all 30 runs, MPS backend; Python 3.12.0, PyTorch 2.14.0, torchvision 0.29.0, scikit-learn 1.9.1.
- **Run record:** each run stores its resolved configuration, commit hash, library versions and hardware, so every row in the results can be traced to the settings in Table 3.

Metrics

- **Macro-F1 (primary):** the mean of the per-class F1 scores,

$$\mathrm{F1}_c = \frac{2\,P_c R_c}{P_c + R_c}, \qquad \text{Macro-F1} = \frac{1}{10}\sum_{c=1}^{10} \mathrm{F1}_c,$$

  where P_c and R_c are the precision and recall of class c. A weak class lowers the score even when overall accuracy is high.

- **Accuracy:** the share of correctly classified images.

- **Cost:**
  - number of trainable parameters;
  - training time, in seconds per epoch and in total;
  - inference latency in milliseconds per image at batch sizes 1 and 128, averaged over 50 timed batches after 5 warm-up batches.

- **Qualitative evidence:** learning curves, the confusion matrix, and correct and incorrect prediction examples for each run.

Seeds, Tuning Budget and Decision Rule

- **Repeated runs:** every reported configuration is trained with seeds 0, 1 and 2 and reported as mean $\pm$ one standard deviation.
- **Equal tuning budget:** at most six configurations per model, compared on validation macro-F1 at seed 0; only the model's own arguments change.
- **Decision rule:** model A is better than model B only if its mean validation macro-F1 is higher *and* the two intervals of mean $\pm$ one standard deviation do not overlap (Figure 14). Otherwise the two are reported as indistinguishable.
- **Accuracy alone never decides a comparison.**

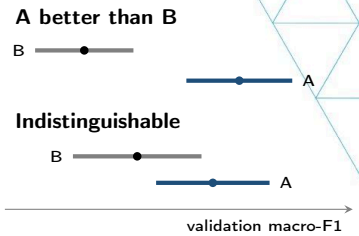


Figure 14: Mean $\pm$ one standard deviation over three seeds.

Experiments and Hypotheses (1/2)

Architecture families

Hypothesis: models with a spatial prior outperform models that read the image as an unordered vector, at comparable capacity.

Varies: model family: Linear, MLP, CNN, GRU, Transformer.

Fixed: the training protocol and the tuning budget.

Metrics: validation macro-F1, accuracy, parameters, seconds per epoch, latency at batch 1 and 128.

Sequence representation

Hypothesis: patch tokens outperform row and column tokens for the recurrent and Transformer models, because a patch keeps neighbouring pixels in both directions together.

Varies: rows, columns, or 4×4 patches.

Fixed: the architecture, its arguments and the protocol.

Metrics: validation macro-F1, against sequence length (28 or 49) and features per token (28 or 16).

Experiments and Hypotheses (2/2)

Recurrent cell

Hypothesis: the GRU matches the LSTM with fewer parameters, because sequences of 28 or 49 steps are short enough that the LSTM's extra gate and cell state add little.

Varies: cell type, GRU (62,218 parameters on rows) or LSTM (82,442).

Fixed: representation, hidden size, depth, dropout and the protocol.

Metrics: validation macro-F1, parameters, seconds per epoch. If the two are indistinguishable, the smaller model is preferred.

A fourth experiment, equalising parameter counts across families, is specified in `EXPERIMENTS.md` as an extension. It was not run for this submission, so no matched-budget result is claimed here.

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Architecture Comparison

Table 4: Mean $\pm$ one standard deviation over seeds 0, 1 and 2. Inference is per image at batch 1; all runs share the machine of Table 3.

Model	Params	s/epoch	ms/img	Val macro-F1	Test macro-F1
CNN	421,642	13.2	0.305	$\mathbf{0.9198 \pm 0.0018}$	$\mathbf{0.9110 \pm 0.0007}$
MLP	535,818	11.3	0.128	0.8862 ± 0.0015	0.8807 ± 0.0018
Transformer	540,170	23.0	1.090	0.8834 ± 0.0023	0.8796 ± 0.0011
GRU	62,218	14.7	2.069	0.8731 ± 0.0012	0.8667 ± 0.0013
Linear	7,850	10.6	0.036	0.7697 ± 0.0025	0.7619 ± 0.0021

- The CNN beats all four; **the MLP and the Transformer are indistinguishable**; both beat the GRU; all four beat the baseline.

Training Behaviour

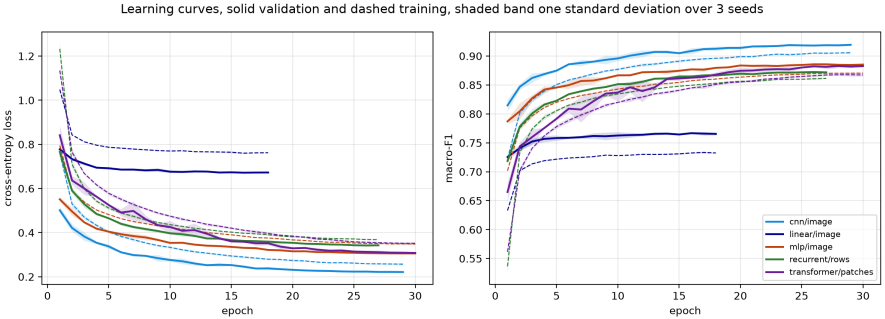


Figure 15: Per architecture, averaged over three seeds.

- Validation scores better than training: augmentation and dropout run in the training pass only, so training is measured on a harder task.
- No model overfits; the baseline underfits, flat near 0.77.

Inductive Bias and Cost

- The CNN wins with **21% fewer parameters than the MLP** and 22% fewer than the Transformer.
- The Transformer is the largest model and the slowest to train (23.0 s/epoch), yet does not separate from an MLP that trains in half the time.
- **CNN:** grid input, locality and translation equivariance; matches the measured pixel correlation, so weight sharing buys accuracy and size at once.
- **MLP:** flat 784-vector, so spatial structure must be learned from data, paid in parameters.
- **GRU:** rows in sequence; real vertical structure, but asserts a processing order a static image does not have. Smallest model with depth, yet slowest at inference (2.069 ms) because its timesteps are sequential.
- **Linear:** 0.7619 measures directly that the classes are not linearly separable in pixel space.

Ablation: Sequence Representation

Table 5: Validation macro-F1 over three seeds, architecture fixed.

Model	Representation	Tokens	Val macro-F1
GRU	rows	28×28	$\mathbf{0.8731 \pm 0.0012}$
GRU	columns	28×28	0.8726 ± 0.0017
GRU	4×4 patches	49×16	0.8215 ± 0.0007
Transformer	rows	28×28	$\mathbf{0.8987 \pm 0.0024}$
Transformer	columns	28×28	0.8961 ± 0.0014
Transformer	4×4 patches	49×16	0.8834 ± 0.0023

- **Hypothesis refuted:** patches are the worst representation for both models, not the best. Patch tokenisation gathers the empty border into a few near-constant tokens.
- The cost is unequal, -0.0516 for the GRU against -0.0153 for the Transformer: attention reaches any token directly, recurrence must cross 75% more steps.

Ablation: Recurrent Cell

Table 6: Row representation fixed; only the gating changes.

Cell	Params	s/epoch	ms/img	Val macro-F1
LSTM	82,442	12.7	0.630	$\mathbf{0.8767 \pm 0.0011}$
GRU	62,218	14.7	2.180	0.8731 ± 0.0012

- **Hypothesis refuted on both counts:** the LSTM is better, by 0.0036 macro-F1 with non-overlapping intervals, *and* cheaper to run despite 33% more parameters.
- Read the timing with care: an LSTM does strictly more arithmetic per step, so the speed is a property of how well each fused kernel is optimised on this backend, not of the algorithms.
- **This revises the main comparison:** on rows the Transformer reaches 0.8987 ± 0.0024 and clears the MLP's 0.8862 ± 0.0015. The tie in Table 4 was an artefact of patch tokenisation, not evidence about attention.

Qualitative Results

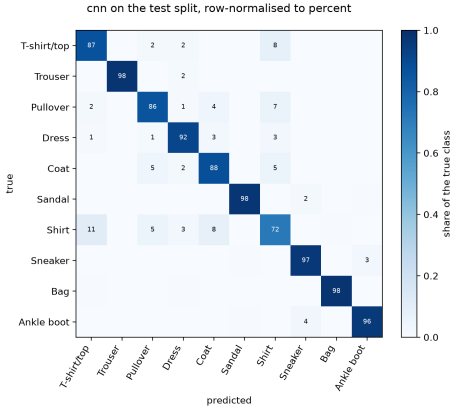


Figure 16: CNN on test, rows normalised to percent: one block of off-diagonal mass.

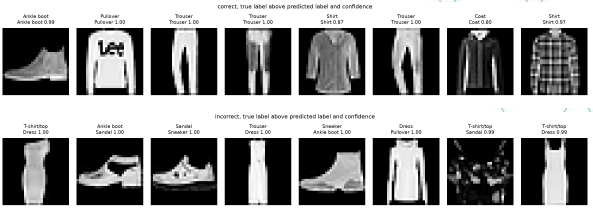


Figure 17: Correct above, the most confident errors below.

Every image in the lower row of Figure 17 fails at confidence 0.99 or above, where the 28×28 image is genuinely compatible with both labels.

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Error Groups

Table 7: CNN on the test split, averaged over three seeds: 888 of 10,000 images, an 8.9% error rate.

Group	Errors	Share	Classes involved
Upper-body garments	576	64.9%	T-shirt/top, Pullover, Coat, Shirt
Footwear	93	10.5%	Sandal, Sneaker, Ankle boot
Scattered	219	24.6%	remainder, 81% of it involves Dress
Total	888	100%	

- **Almost two thirds of all errors sit in one four-class block**, and every confusion inside it is bidirectional (Table 7).
- Largest flows: Shirt → T-shirt/top 118 of 1,000 Shirts, T-shirt/top → Shirt 78, Shirt → Coat 75.
- **Shirt is the floor for all five models**: F1 0.7365, precision 0.753, recall 0.721. It is the only class below 0.80.

Causes and Improvements

- **The cause is in the data, not the model.** These four classes differ by sleeve length, collar shape and fabric texture, cues that survive 28×28 grayscale poorly. The EDA predicted it: they have the most similar class means and dominate a nearest-class-mean classifier before any network is trained.
- **It is not architectural.** The same block holds 64% of the MLP's errors, 64% of the GRU's, 61% of the Transformer's and 48% of the baseline's. The baseline is the exception that confirms the rule: its errors are least concentrated only because it makes 2,369 of them.
- **Footwear is not worth attacking:** sole shape and ankle height survive downsampling, so all three exceed 0.95 F1 on only 93 errors.

Capacity is demonstrably not the fix

The Transformer carries 28% more parameters than the CNN and a *worse* Shirt F1. Three directions match the diagnosis instead: a class-weighted or focal loss trading precision on the easier classes for Shirt recall; augmentation targeted at sleeve and collar geometry rather than a generic crop; or a second-stage classifier specialising on the four-class block.

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Findings

Research question: does inductive bias or capacity govern performance here?

The evidence favours inductive bias.

- The most accurate model is the *smallest* of the three strong ones, and keeps its margin even over the best sequence configuration the ablations found.
- The largest model, trained at twice the cost per epoch, is indistinguishable from a cheaper one.
- The error structure that dominates every model's residual was predicted from the class means before any network was trained.
- **Selected model: the CNN**, 0.9110 ± 0.0007 macro-F1 on test, and also the cheapest to run among the models competitive on accuracy, so there is no accuracy-against-speed trade-off to negotiate here.
- **A result can hinge on a choice made before any measurement:** the Transformer's apparent tie with the MLP disappeared once the representation was varied.

Limitations and Future Work

Limits

- One dataset of 28×28 grayscale images with balanced classes; neither the ranking nor the error structure should be assumed to transfer.
- Three seeds estimate a deviation from three observations, so a tie means *not separated at this sample size*, not equal.
- One shared protocol and an equal search cap, so each figure is a score under a common budget, not the best each architecture could reach.
- No pretrained weights, which constrains the Transformer most.

Next

- Ablate the representation for the remaining three architectures; only two of the five were tested at more than one view of their input.
- Reweight the loss towards Shirt, the single class that bounds macro-F1 for all five models.
- Equalise parameter counts across families to separate capacity from architecture directly.

Thank You

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